LESSON 6.6

Identify Equivalent Algebraic Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can show that two expressions are equivalent by simplifying and combining like terms.

WHY THIS MATTERS At the music studio, two producers wrote formulas for the cost of renting equipment. Producer A wrote 3x + 2x + 10, and Producer B wrote 5x + 10. They look different, but are they really? Let's check: if x = 4, Producer A gets 3(4) + 2(4) + 10 = 30, and Producer B gets 5(4) + 10 = 30. Same result — they're equivalent expressions!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equivalent Expressions *Spanish: Expresiones equivalentes*	Expressions that look different but have the same value for every allowed variable value.	7c + 9 + 6c = 13c + 9
Equivalent *Spanish: Equivalente*	Expressions that always have the same value.	2x + 4 and 2(x + 2): when x = 3, both = 10; when x = 7, both = 18 — always the same
Simplify *Spanish: Simplificar*	To rewrite an expression in a shorter form that is still equal to the original.	7c + 9 + 6c → 13c + 9
Like Terms *Spanish: Términos semejantes*	Terms with the same letter raised to the same power, like 2x and 5x.	7c and 6c
Combine *Spanish: Combinar*	To add or subtract terms with the same letter into one.	7c + 6c = 13c
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	the 13 in 13c

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Simplify 7c + 9 + 6c. Is it equivalent to 13c + 9?

1. Like terms: 7c and 6c.
2. Combine: 7c + 6c = 13c. The 9 stays alone.
3. 7c + 9 + 6c = 13c + 9.

Answer: 13c + 9; yes, equivalent

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's check 2x + 3x and 5x. Try x = 2.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The like terms are _____ and _____ because _____ .
- I combined them to get _____ .
- The constant _____ stays by itself because _____ .
- The two expressions _____ equivalent because _____ .

Word bank equivalent | like terms | combine | simplify | coefficient | constant | term | variable | same value | group | add | compare

WATCH OUT!

A common mistake

A common mistake in Equivalent Expressions is adding the exponents instead of the coefficients when combining like terms

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Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to 4x + 3x?

(A) 7x
(B) $7x^2$
(C) 12x
(D) 43x

STEPS TO TRY

1. 4x and 3x share the exact same variable, x — that makes them like terms.
2. Combine like terms by adding the coefficients (the number parts). The variable tags along unchanged.
3. Add the coefficients: 4 + 3. Keep the x exactly as it is — no new exponent appears.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which expression is equivalent to 2(m + 5)?

(A) 2m + 10
(B) 2m + 5
(C) 7m
(D) m + 10

STEPS TO TRY

1. The 2 outside the parentheses must multiply both the m and the 5.
2. Distribute: write 2 × m and 2 × 5, then join the products with +.
3. Start: 2 × m = 2m. Now multiply 2 × 5 and add that product on.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

Which property is shown: 3(x + 7) = 3x + 21?

(A) Commutative
(B) Distributive
(C) Associative
(D) Identity

STEPS TO TRY

1. Watch what happened to the 3: it multiplied the x AND the 7 across the parentheses.
2. The order didn't change and the grouping didn't just move — a factor was handed out to each term. Which property does that?
3. Check: 3 × x = 3x and 3 × 7 = 21 — both appear on the right. Name the property that multiplies across a sum.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Simplify: 5n + 2n

(A) 7n

(B) $7n^2$

(C) 10n

(D) 52n

STEPS TO TRY

1. 5n and 2n both have the variable n — they're like terms, so they can combine.
2. Add the number parts (coefficients) and keep n the same. Don't multiply them.
3. Compute 5 + 2 for the new coefficient, then write it in front of n.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you decide which expressions belong in the same group? What strategies did you use?

SENTENCE STARTER I knew ___ and ___ are equivalent because when I ___ the terms, I got the same expression.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help